

```
7. prices = [price]
8.
9. for _ in range(days-1):
10.     # Random price change with slight upward bias
11.     price += np.random.normal(0.05, 1)
12.     prices.append(price)
13.
14. # Plot the stock price
15. plt.figure(figsize=(10, 6))
16. plt.plot(prices)
17. plt.title('Simulated Stock Price')
18. plt.xlabel('Days')
19. plt.ylabel('Price')
20. plt.grid(True)
21. plt.show()
```

Additional Essential Libraries

While NumPy, pandas, and Matplotlib form the core toolkit, several other libraries are valuable for algorithmic trading:

1.

scikit-learn: A machine learning library used for predictive modeling

```
1. pip install scikit-learn
```

2.

statsmodels: Statistical modeling for time series analysis

```
1. pip install statsmodels
```

3.

yfinance: A popular library for downloading financial data from Yahoo Finance

```
1. pip install yfinance
```